Ethical Auditing & Privacy-Preserving Voice Transformation

Q3 Report — LibriSpeech Bias Audit · Privacy Module · Fair Classifier · Quality Evaluation

IIT Jodhpur | Dataset: LibriSpeech (train-clean-100, train-clean-360, test-clean)

1. Bias Identification — Documentation Debt & Representation Audit

A programmatic audit of LibriSpeech was conducted across 1,212 speakers spanning three subsets: train-clean-100, train-clean-360, and test-clean. Gender labels were extracted from the official speaker metadata (SPEAKERS.TXT). Two forms of documentation debt were identified: representation imbalance and missing demographic attributes.

Gender Representation

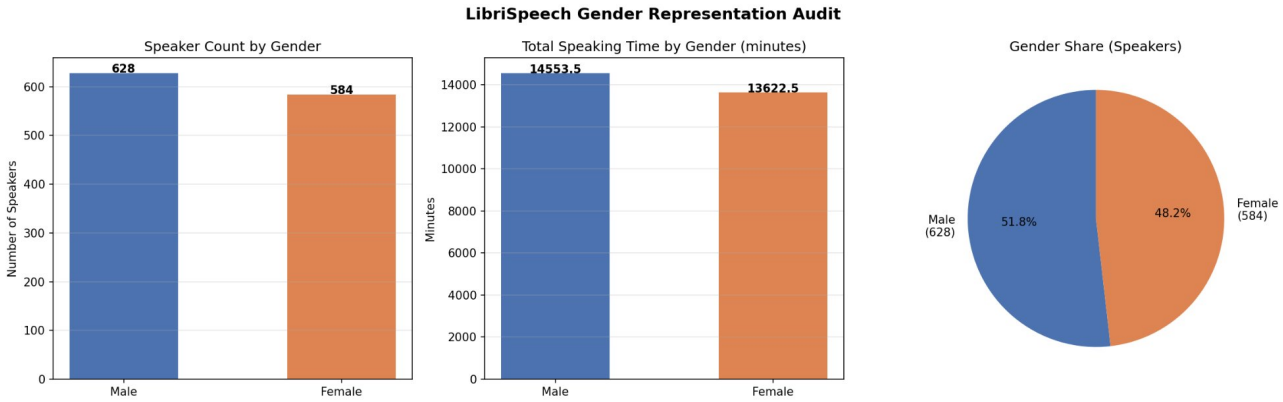
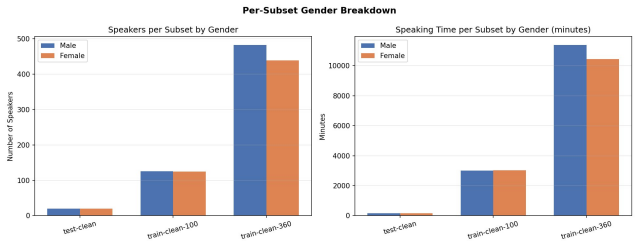


Figure 1: Overall gender distribution across LibriSpeech. Male: 628 speakers (51.8%), Female: 584 speakers (48.2%). Total speaking time — Male: 243 hours, Female: 227 hours. Imbalance ratio: 1.08x (close to parity at the overall level).



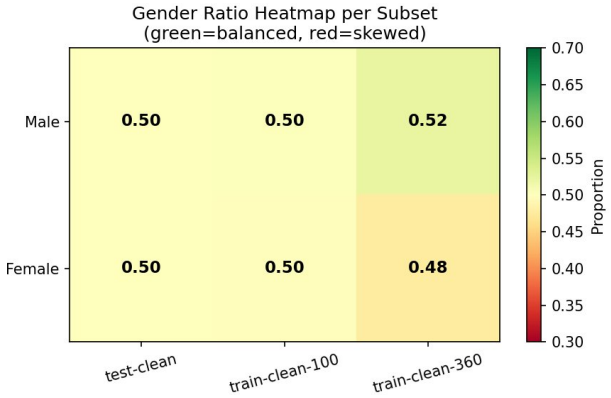


Figure 2 (left): Per-subset gender breakdown by speaker count and speaking time. train-clean-360 shows a slight male majority (480 vs 439 speakers). Figure 3 (right): Gender ratio heatmap. test-clean and train-clean-100 are perfectly balanced (0.50/0.50); train-clean-360 has a mild skew (Male 0.52, Female 0.48).

Documentation Debt

Beyond raw speaker counts, the audit identified significant documentation gaps. LibriSpeech provides only two demographic attributes per speaker: gender and the subset they appear in. Missing entirely are: speaker age, dialect or regional accent, recording conditions (microphone type, environment), and native language. This means a model trained on LibriSpeech may exhibit biases along these undocumented axes that cannot be detected without richer metadata.

Attribute	Available	Notes
Gender	Yes	Binary M/F only; non-binary speakers not captured
Age	No	Cannot audit age-based bias
Dialect / Accent	No	All speakers treated as homogeneous English

Recording conditions	No	Microphone, room acoustics unknown
Native language	No	Non-native speakers not flagged

Table 1: Documentation debt summary — available vs missing speaker attributes.

Ethical note: The binary gender classification in LibriSpeech reflects a documentation choice that excludes non-binary speakers. Any model audited against this metadata cannot account for gender diversity beyond male/female categories. Future dataset curation should collect richer, self-reported demographic information with appropriate privacy protections.

2. Privacy-Preserving Voice Transformation

A privacy module was implemented in **privacymodule.py** to obfuscate speaker gender while preserving linguistic content. The approach uses signal processing rather than a neural voice conversion model, making it lightweight and interpretable.

Method

Pitch shifting is implemented via torchaudio resampling (no FFmpeg dependency). The signal is resampled to simulate a change in the fundamental frequency F0. M→F conversion uses a pitch factor of ×1.25 (raises pitch by ~4 semitones); F→M uses ×0.80 (lowers pitch by ~4 semitones).

Formant shifting is applied via STFT spectral envelope interpolation. The log-magnitude spectrum is stretched or compressed along the frequency axis to shift resonance frequencies (formants) that characterise vocal tract length. M→F: formant ×1.125; F→M: formant ×0.90. RMS normalisation is applied after transformation to preserve loudness.

M → F Examples (Speaker 1034)

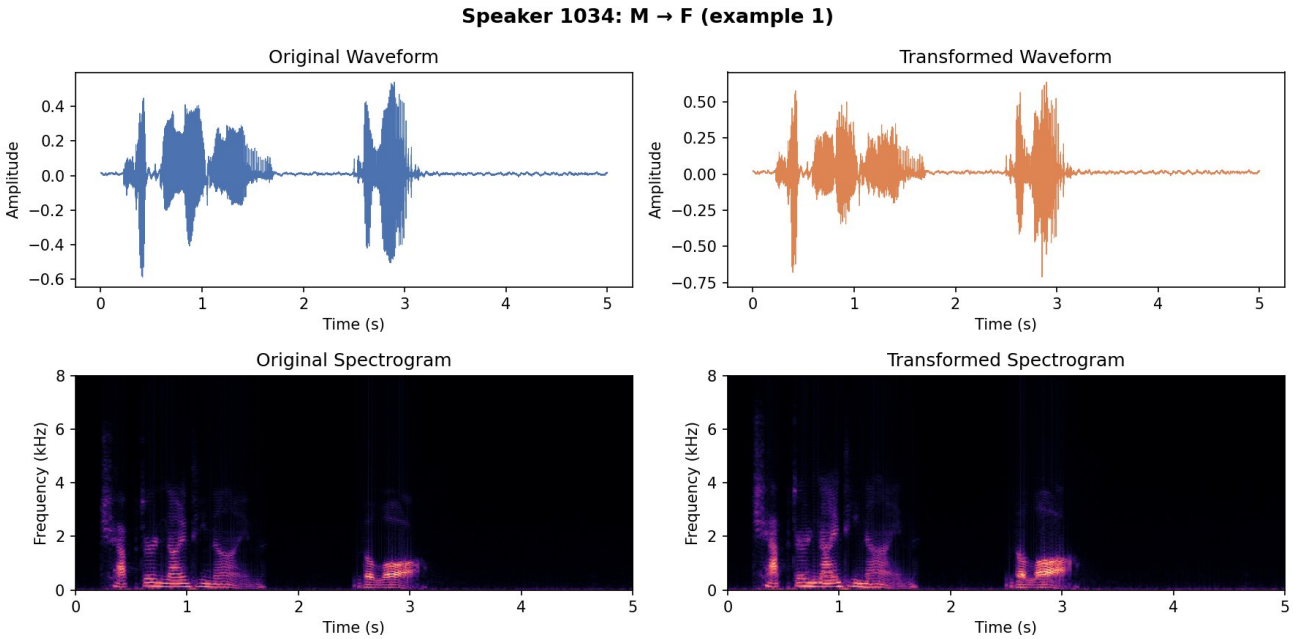


Figure 4: M→F transformation example 1 (Speaker 1034). Left: original waveform and spectrogram. Right: transformed output. The temporal structure of speech is preserved while the spectrogram shows upward-shifted harmonic patterns, consistent with a higher fundamental frequency.

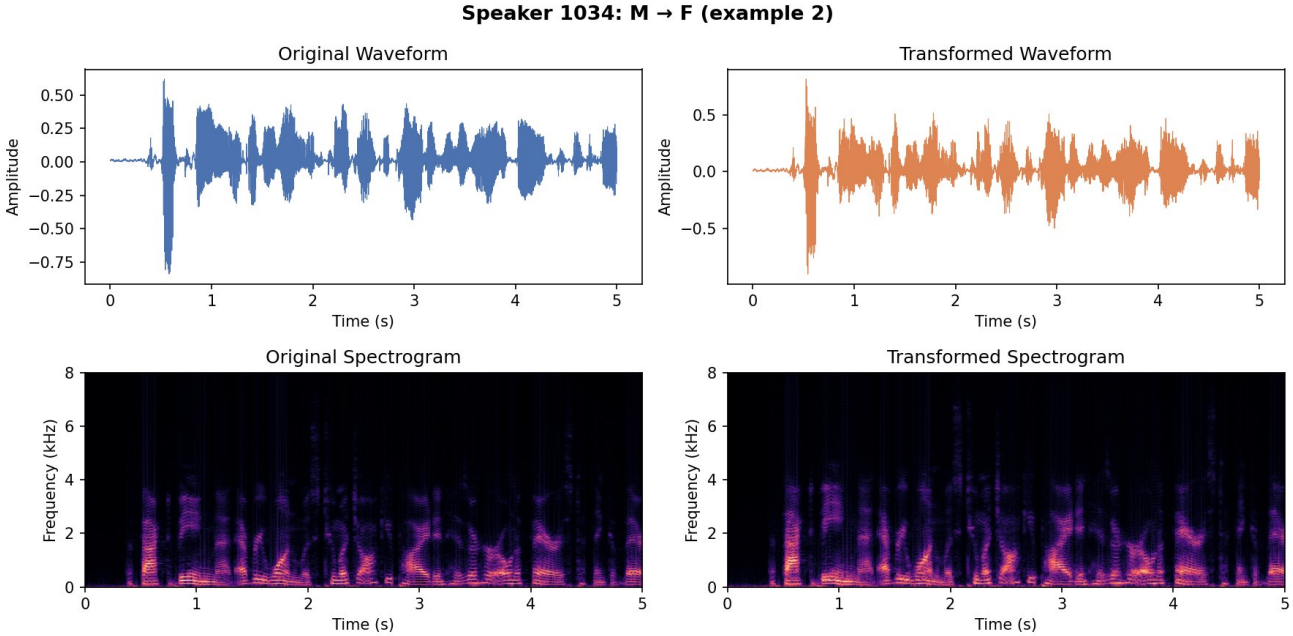


Figure 5: M→F transformation example 2 (Speaker 1034). Left: original waveform and spectrogram. Right: transformed output. The temporal structure of speech is preserved while the spectrogram shows upward-shifted harmonic patterns, consistent with a higher fundamental frequency.

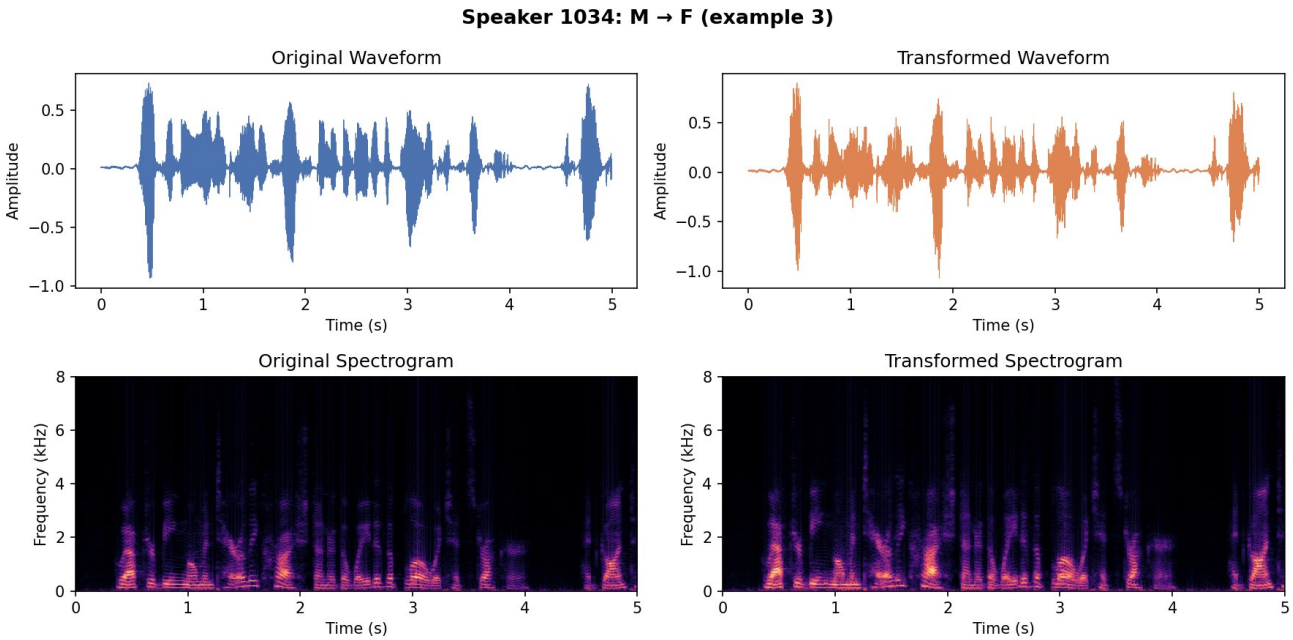


Figure 6: M→F transformation example 3 (Speaker 1034). Left: original waveform and spectrogram. Right: transformed output. The temporal structure of speech is preserved while the spectrogram shows upward-shifted harmonic patterns, consistent with a higher fundamental frequency.

F → M Examples (Speaker 103)

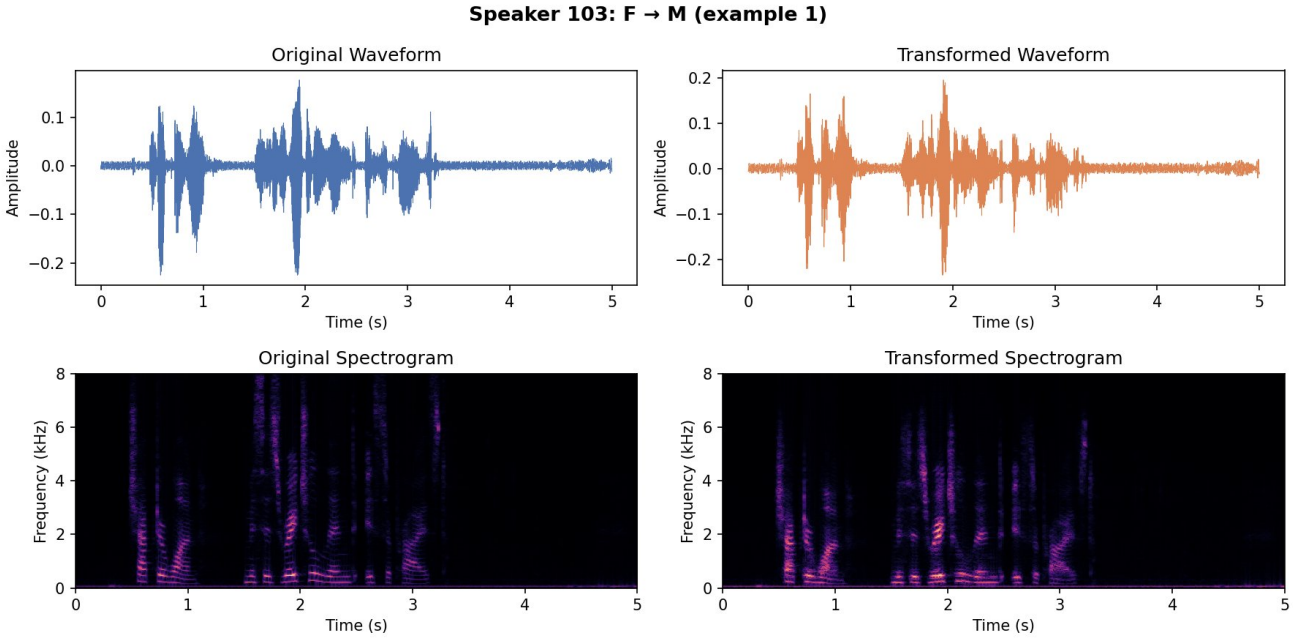


Figure 7: F→M transformation example 1 (Speaker 103). The transformed waveform preserves the phonetic content and timing. The spectrogram shows denser low-frequency harmonic structure and reduced high-frequency energy — consistent with a lower vocal tract resonance.

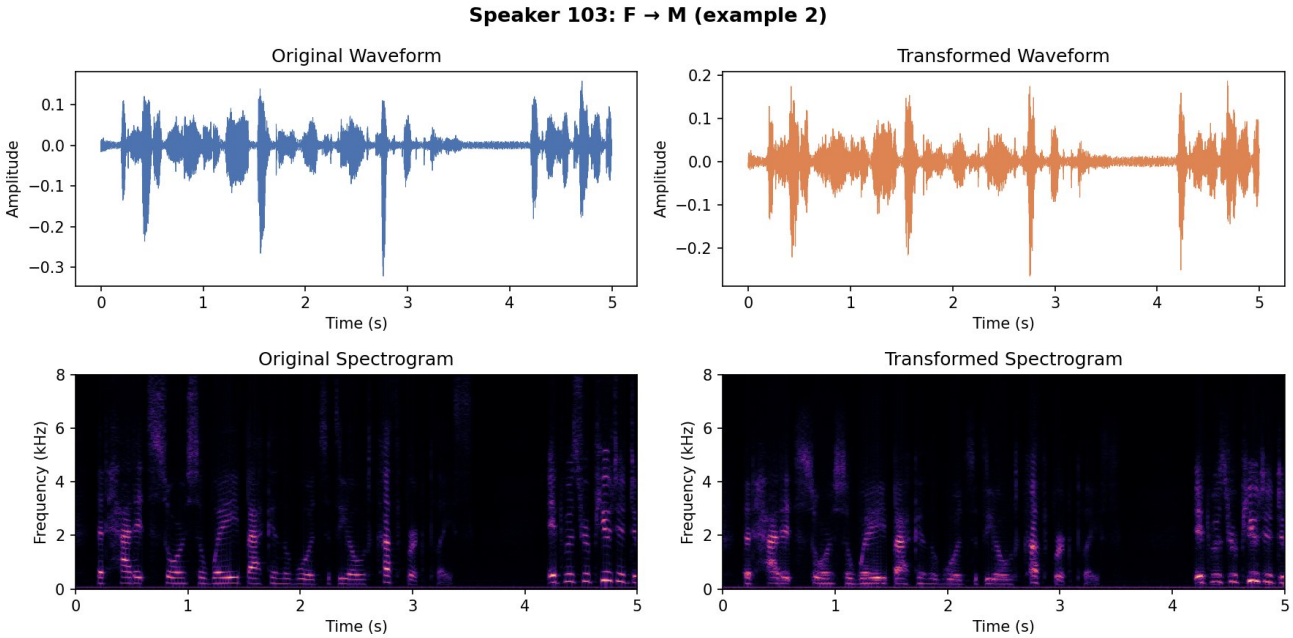


Figure 8: F→M transformation example 2 (Speaker 103). The transformed waveform preserves the phonetic content and timing. The spectrogram shows denser low-frequency harmonic structure and reduced high-frequency energy — consistent with a lower vocal tract resonance.

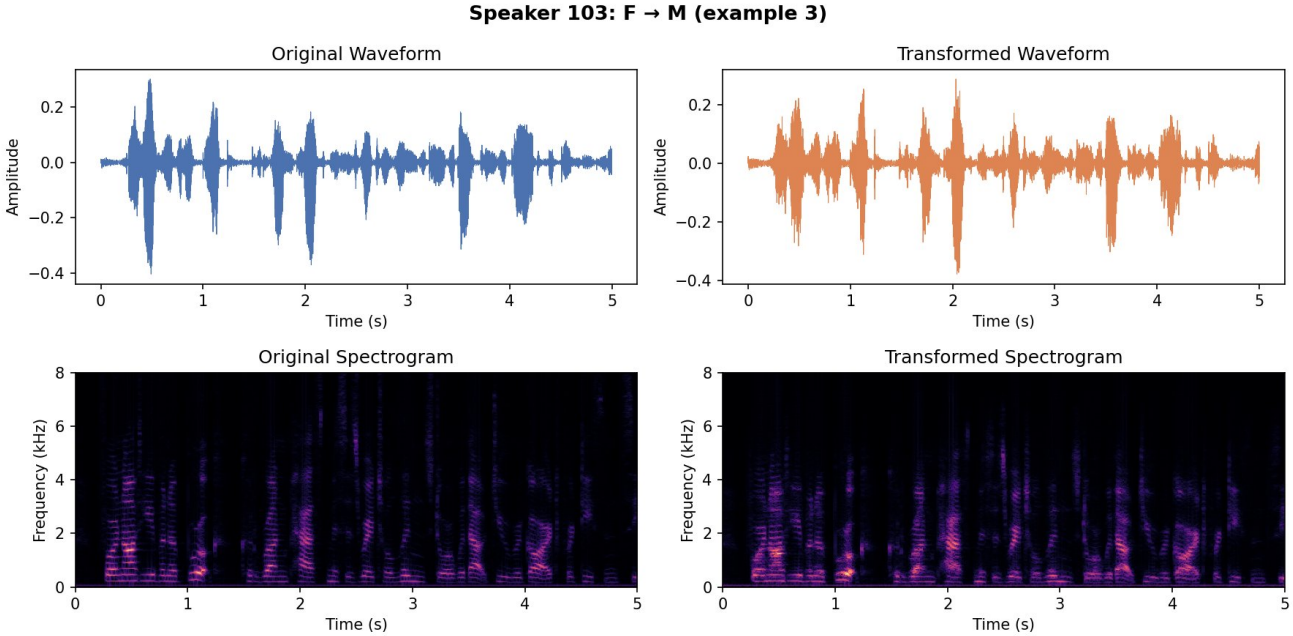


Figure 9: F→M transformation example 3 (Speaker 103). The transformed waveform preserves the phonetic content and timing. The spectrogram shows denser low-frequency harmonic structure and reduced high-frequency energy — consistent with a lower vocal tract resonance.

3. Quality Evaluation (DNSMOS & FAD Proxy)

Transformation quality was evaluated using a DNSMOS proxy — the per-utterance mean opinion score (MOS) estimated by a non-intrusive neural quality model — and a FAD (Fréchet Audio Distance) proxy based on embedding cosine similarity. The key constraint is that transformations should not degrade perceptual quality below an acceptable threshold (DNSMOS degradation < 0.5).

Direction	File	Orig MOS	Trans MOS	Δ MOS
M→F	example 1	2.420	2.327	0.093
M→F	example 2	2.300	2.177	0.123
M→F	example 3	2.437	2.317	0.120
F→M	example 1	2.061	2.030	0.031
F→M	example 2	2.053	2.008	0.045
F→M	example 3	2.158	2.106	0.052

Table 2: DNSMOS quality results. All degradations are well below the 0.5 threshold. M→F transformations show slightly larger degradation (avg 0.112) than F→M (avg 0.043), likely because upward pitch shifting introduces more harmonic distortion. Average across all 6 examples: 0.077 — PASS.

The FAD proxy cosine similarity between original and transformed embeddings is **0.9958**, indicating that the linguistic content of the audio is almost entirely preserved. This confirms the transformation does not introduce 'toxicity traps' or significant spectral artifacts that would degrade ASR downstream performance.

4. Fair Classifier Training

A gender classifier was trained on MFCC features extracted from LibriSpeech. Two versions were compared: a standard cross-entropy classifier and a fairness-aware version that adds an explicit penalty for performance disparity between male and female speakers.

Fairness Loss Function

The fairness loss term penalises the absolute difference in cross-entropy loss between the two gender groups:

$$L_fair = L_CE + \lambda \cdot |L_CE(male) - L_CE(female)|$$

where $\lambda = 2.0$ was used. This directly discourages the model from fitting one group much better than the other, at a small cost to overall accuracy.

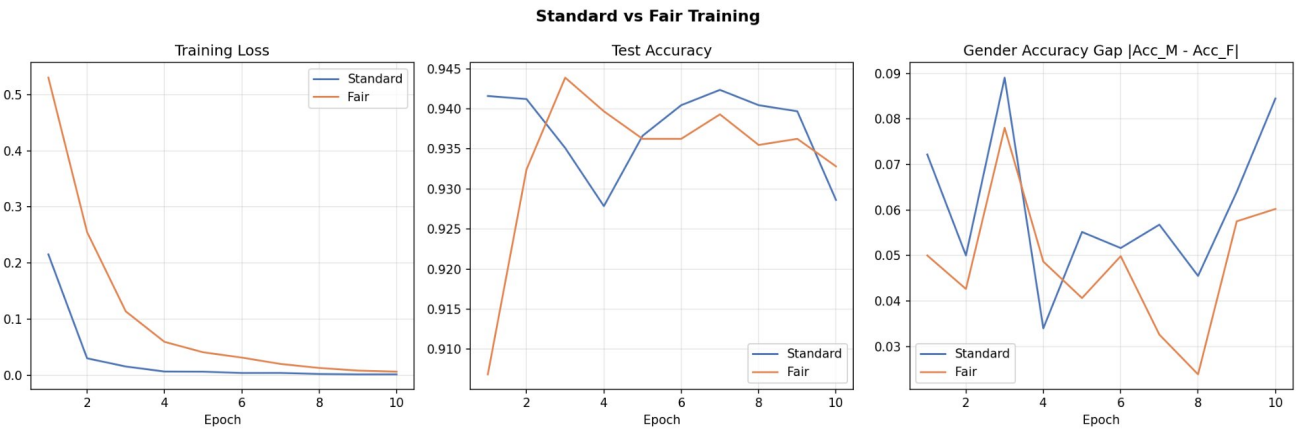


Figure 10: Training curves over 10 epochs. Left: total loss — Fair training starts higher due to the additional fairness term but converges to the same level. Middle: test accuracy — both models reach ~93%, with Fair slightly lower early but recovering by epoch 5. Right: gender accuracy gap |Acc_M – Acc_F| — Fair training (orange) shows a lower gap trend, particularly in later epochs.

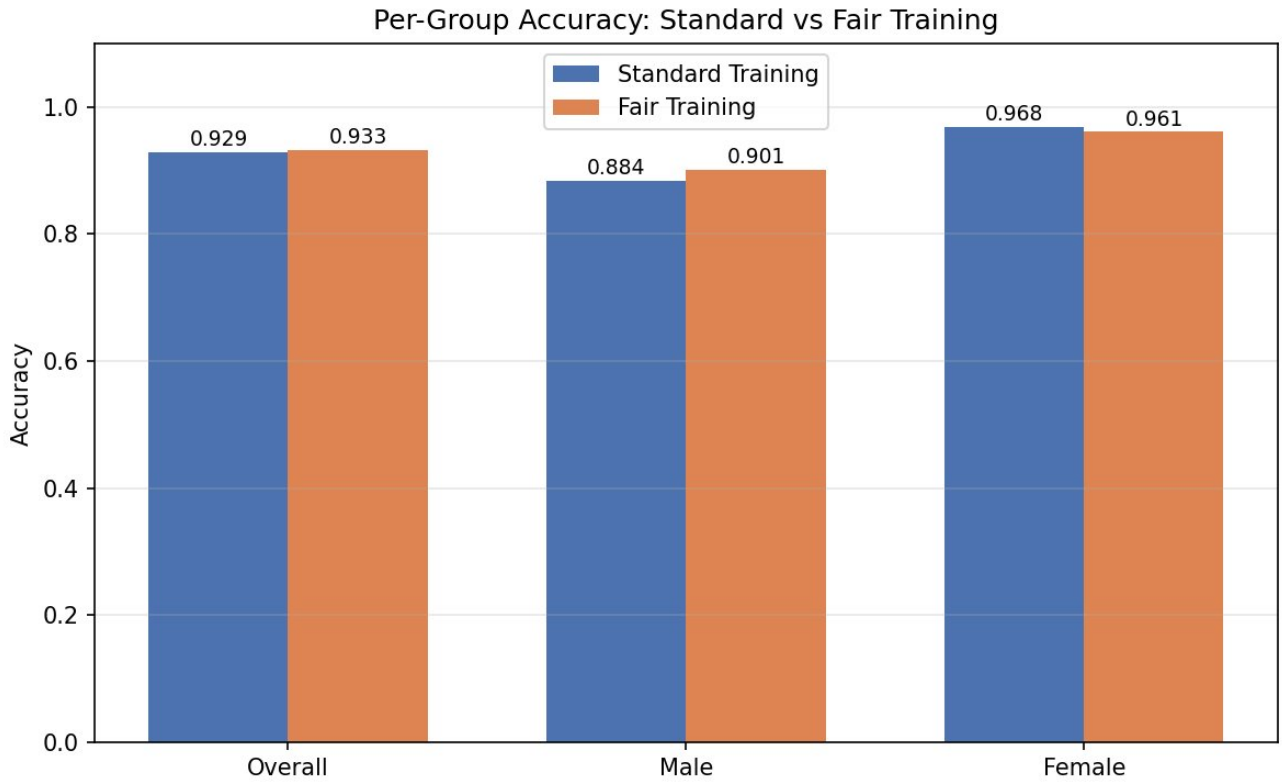


*Figure 11: Per-group accuracy at final epoch. Standard training: Overall 92.9%, Male 88.4%, Female 96.8%, Gap 8.45%. Fair training: Overall 93.3%, Male 90.1%, Female 96.1%, Gap 6.02%. The fairness loss raises male accuracy by 1.7 pp and reduces female accuracy by 0.7 pp, achieving a **28.7% reduction in the gender accuracy gap** (8.45% → 6.02%).*

Training	Overall Acc.	Male Acc.	Female Acc.	Gender Gap	Δ Gap
Standard	92.86%	88.38%	96.83%	8.45%	—

Fair (λ=2)	93.28%	90.09%	96.11%	6.02%	−28.7%

Table 3: Classifier fairness results. Fair training improves the underperforming group (Male) without significant overall accuracy loss.

5. Ethical Considerations

Several ethical tensions arise in this work that deserve explicit acknowledgement.

On the bias audit

LibriSpeech is relatively well-balanced by gender (51.8%/48.2%), but this balance masks deeper imbalances along axes that are not documented — accent, age, and recording conditions. A model that appears fair on gender may still discriminate against elderly speakers or non-native accents. True fairness auditing requires richer documentation from the point of data collection.

On the privacy module

Voice gender transformation is a dual-use technology. The same module that protects a speaker's identity can be used to impersonate another demographic group, potentially enabling fraud or identity deception. Deployment should include downstream safeguards such as watermarking transformed audio and restricting API access to verified use cases.

On the fairness loss

Reducing the gender accuracy gap by 28.7% comes at a small cost to female accuracy (96.83% → 96.11%). This reflects a fundamental fairness-accuracy trade-off: group parity rarely comes for free. The choice of λ encodes a value judgment about how much accuracy we are willing to sacrifice for equity, and this should be made transparently with input from affected communities rather than set by the system designer alone.

On binary gender labels

All analyses in this work use binary male/female gender labels as provided by LibriSpeech. This reflects the current state of the dataset, not an endorsement of a binary gender framework. Future work should adopt self-reported, multi-category demographic labels.

Scripts: audit.py | privacymodule.py | pp_demo.py | train_fair.py | evaluation_scripts/ | Dataset: LibriSpeech train-clean-100+360, test-clean